KAUSTUBH DIXIT

Software Development Engineer - III

CONTACT

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EDUCATION

SRM Institue of Technology

2016 - 2020 I **CGPA: 91.6%**
BTech in Computer Science

Udacity

2019 I Data Scientist Nanodegree

City Montessori School, Lucknow

2015 I **Percentage: 95.3%**
ISC Board (XII)

2013 I **Percentage: 93.5%**
ISC Board (X)

PROJECTS

Segmentation of Nerves in Ultrasound Images

Image segmentation of ultrasound images to mitigate pain (GitHub)

- Constructed custom image transformers and custom training and validation generators to feed the data to the model.
- Trained a U-Net encoder-decoder network for image segmentation using Keras.

Automated door unlocker using ML and Arduino

Built an automated door unlocker using Arduino and Convolutional Neural Nets (GitHub)

- Leveraged OpenCV library to encode the vector representation of images.
- Utilised dlib models (pre-trained on 3 million+ images) to recognise facial features for detection.
- Coupled ML model with Arduino to drive the door-unlocking mechanism

Natural Disaster Relief using Text Classification

Categorizing the texts of disaster-struck people using NLP (GitHub)

- Created an ETL pipeline to pre-process the text, including tokenisation, stemming and lemmatisation.
- Trained an SVC (Support Vector Classifier) on this data using the sklearn library to correctly classify the text.
- Utilised the Flask web framework to create APIs for deploying the web app

SKILLS

Frontend: React, Angular, HTML, CSS, JavaScript, Typescript, SCSS

Backend: Python, AWS (Lambda, S3, SNS, SQS, RDS, API Gateway, DynamoDB, RDS), MongoDB, PostgreSQL, FastAPI, Docker, GCP

ML/AI: PyTorch, Keras, Computer Vision, NLP, scikit-learn, SpaCy, OpenAI, Gemini, GenAI, Seaborn, Tableau

EXPERIENCE

SDE-3 | Teachmint

August 2025 – Present

Engineered core backend systems for an AI-driven educational platform delivering intelligent content and learning tools to students and teachers.

- Built microservices using AWS Lambda and FastAPI to add core functions and modules for the platform team.
- Developed a production-grade payment and subscription system using Razorpay, handling webhooks, reconciliation and secure transaction flows at scale for lakhs of students.
- Architected an end-to-end, multi-agent AI video generation pipeline using Gemini API and LangGraph, orchestrating content generation, evaluation, retries and final delivery for educational content.
- Refactored non-performant python codebases, improving VM resource utilization and reducing costs by $2,000+ per month.
- Deployed an AI-powered mindmap generation feature via OpenAI API that transformed NCERT content into hierarchical visual summaries for students.

SDE-2 | Perspect AI

April 2022 – July 2025

Automated employee hiring and skill matching using psychometric assessments and data-driven evaluation pipelines.

- Engineered critical gaming modules using AWS serverless architecture, scaling to 20k+ concurrent users.
- Collaborated with psychology researchers to develop scientifically valid and robust assessment models.
- Led AI/ML initiatives for an in-house face recognition system in PyTorch, reducing operational costs by ~₹1 lakh per month.
- Built a coding assessment platform evaluating proficiency across 10+ programming languages.
- Designed and deployed an end-to-end RAG pipeline using Pinecone vector DB and OpenAI embeddings for context-grounded LLM responses over the internal knowledge base.
- Owned backend architecture for AI-first features, ensuring scalability, fault tolerance and observability in production systems..

Machine Learning Engineer (Team Lead) | Taiyo (Remote)

June 2019 – March 2022

Led a team of 4 to build an automated trading system focused on risk and uncertainty management.

- Developed frontend experiences using React and Angular with efficient state management and routing.
- Optimized page load times by ~12 seconds using caching, lazy loading, and bundle analysis.
- Built REST APIs using Django and Dockerized microservices.
- Achieved 30× speedups in feature generation using multiprocessing and asynchronous pipelines.
- Deployed classical and deep learning models for asset price prediction using macro and microeconomic data.